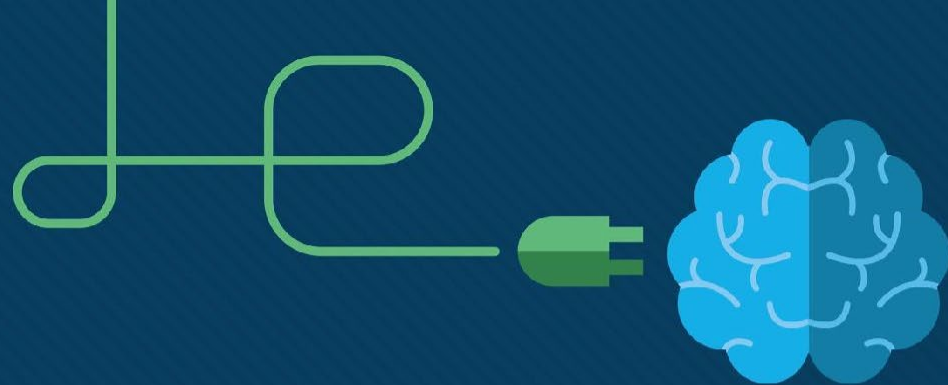




Week 11: Network Security Devices & ACLs



Understanding Network Security Devices

- Network security devices
 - Router Access Control List (ACL)
 - Firewalls/ WAF
 - Intrusion detection/prevention systems
 - EDR/XDR
 - SIEM/SOAR
 - ...

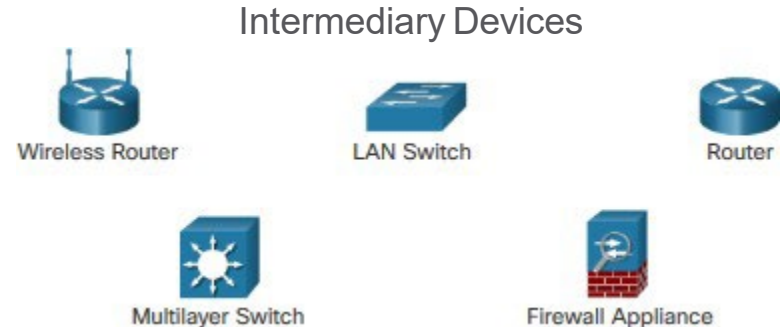
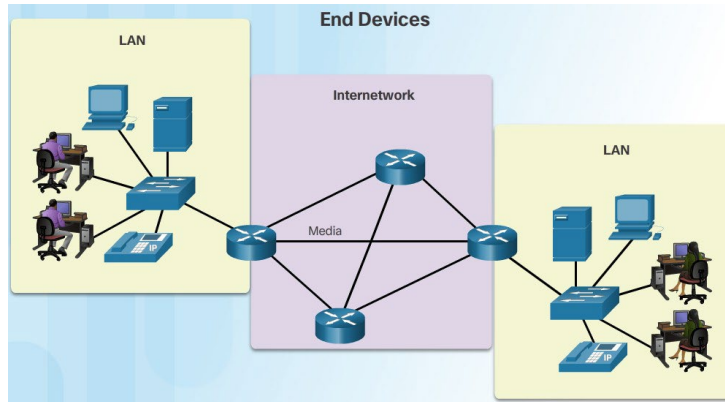
Network Devices

▪ End Devices:

- Computers, laptops, servers, printers, smart devices, and mobile devices.
- Individual end devices are connected to the network by intermediary devices.

▪ Intermediary Devices:

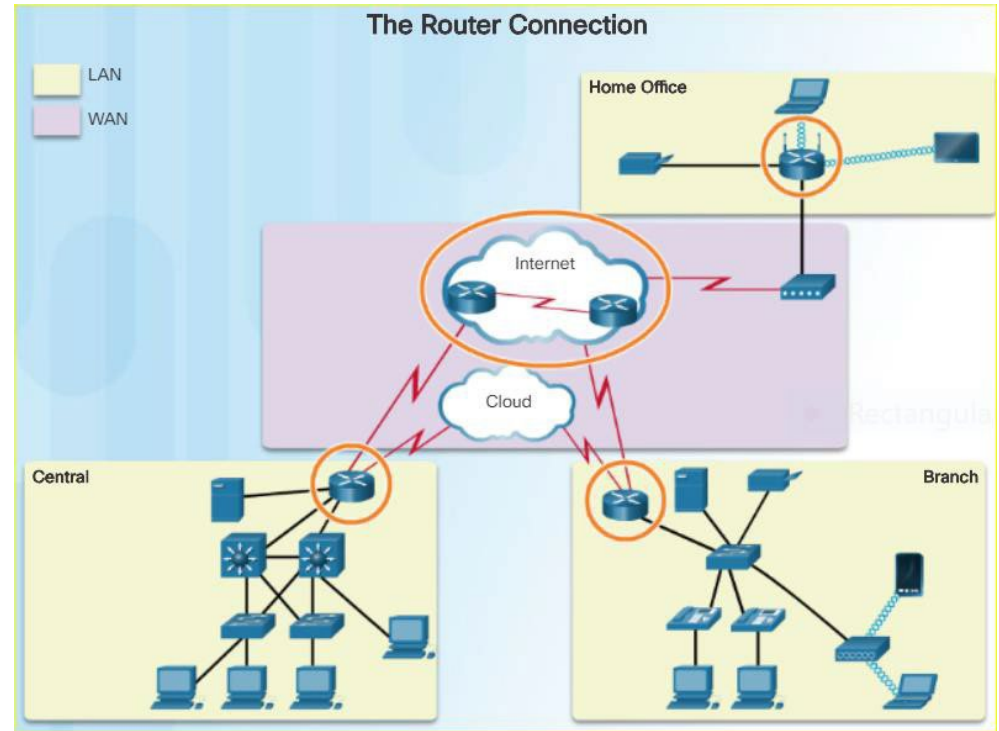
- Connect the individual end devices to the network and also connect multiple individual networks to form an internetwork.
- Provide connectivity and ensure that data flows across the network.



Network Devices

Routers

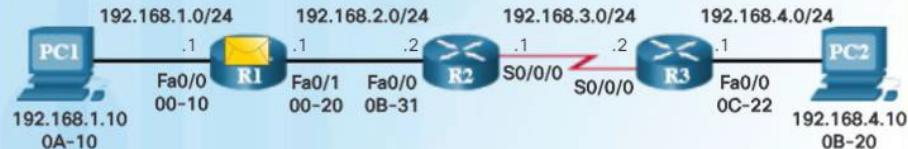
- Function of a Router:
 - Provides path determination and packet forwarding.
 - Responsible for encapsulating and de-encapsulating packets.
 - Uses a routing table to determine the best path to use to send packets to a specified network.
- Routing Table:
 - Contains directly connected routes and remote routes.
 - Router searches its routing table for a network address that matches the destination IP address of a packet.
 - Uses the gateway of last resort if learned or configured; otherwise, the packet is discarded.



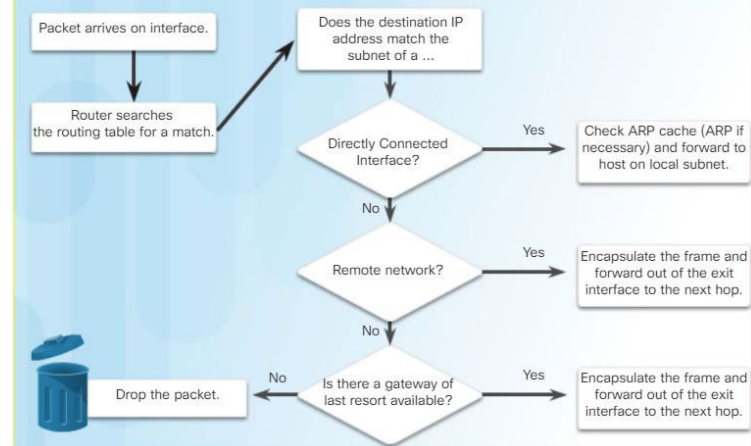
Router Operation

- A primary function of a router is to determine the best path to use to send packets to each subnet. To determine the best path, the router searches its routing table for a network address that matches the destination IP address of the packet. The routing table search results in one of three path determinations:
 - **Directly connected network**
 - **Remote network**
 - **No route determined**

PC1 Builds a Packet to Send to PC2

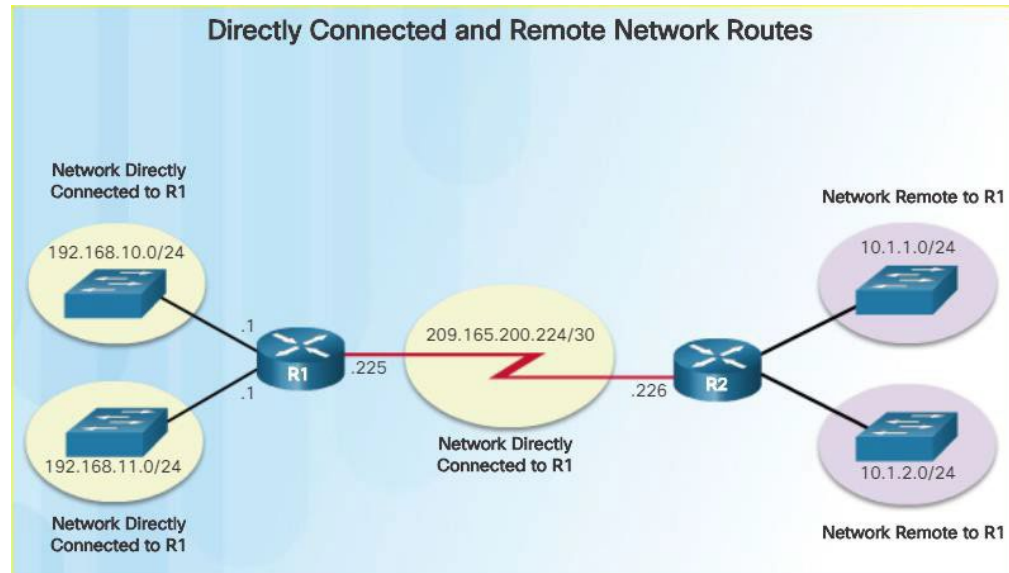


Packet Forwarding Decision Process

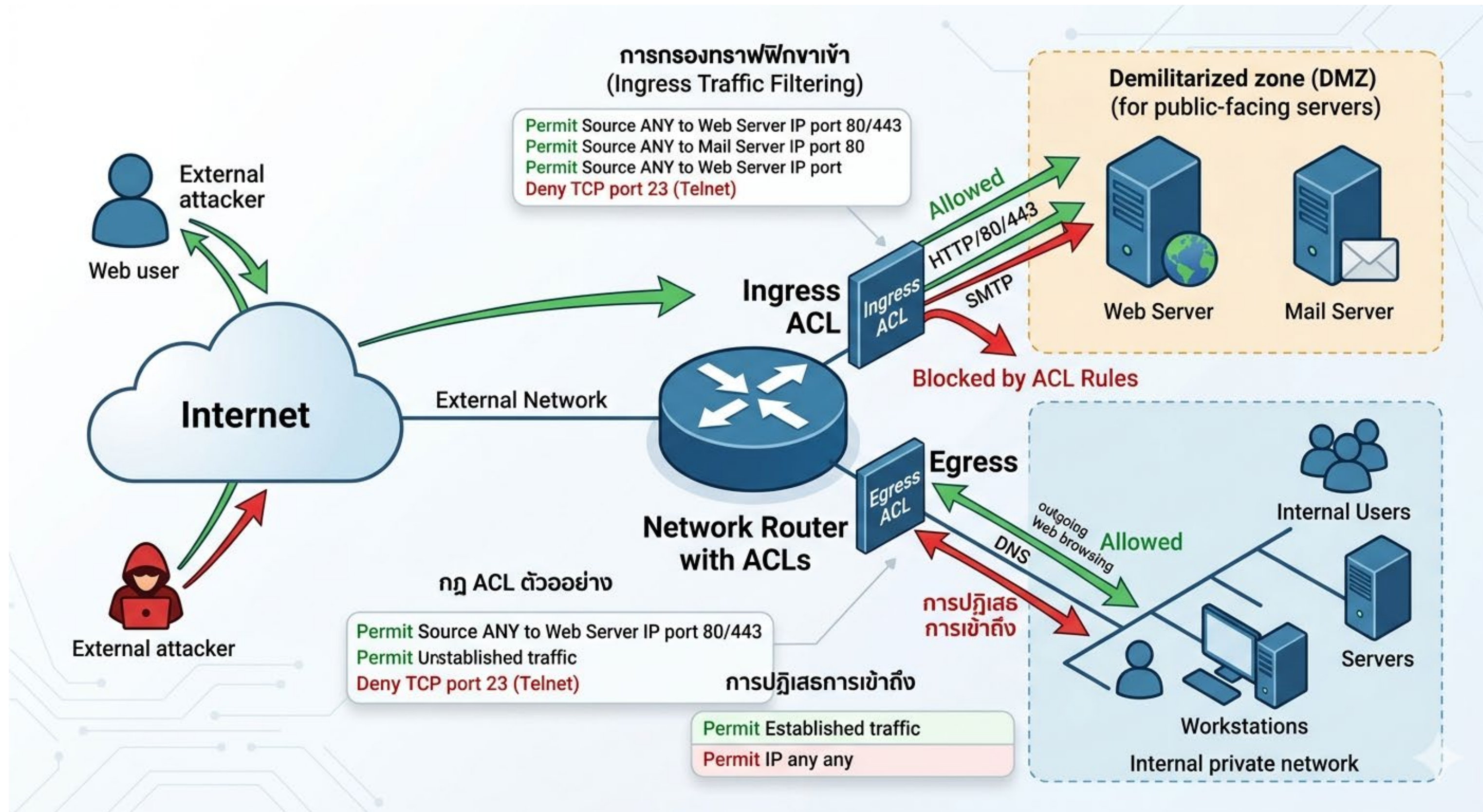


Routing Information

- The routing table of a router stores the following information:
 - **Directly connected routes**
 - **Remote routes**
- The destination network entries in the routing table can be added in several ways:
 - **Local Route interfaces** – These are added when an interface is configured and active.
 - **Directly connected interfaces** – These are added to the routing table when an interface is configured and active.
 - **Static routes** – These are added when a route is manually configured and the exit interface is active.
 - **Dynamic routing protocol** – This is added when routing protocols that dynamically learn about the network, such as EIGRP or OSPF, are implemented and networks are identified.



Router Access Control List (ACL)



11.1 Network Security Infrastructure

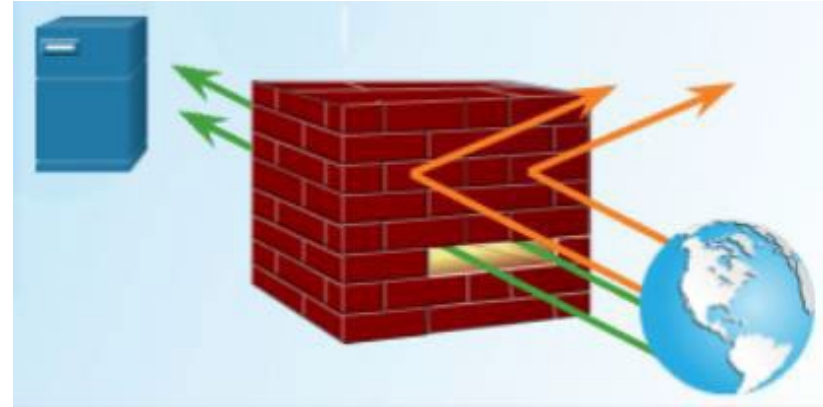
Understanding Firewalls

- Firewalls are hardware devices or software installed on a system and have two purposes
 - Controlling access to all traffic that enters an internal network
 - Controlling all traffic that leaves an internal network
- Advantages of hardware firewalls
 - They are usually faster than software firewalls
 - They can handle a larger throughput than software firewalls

Security Devices

Firewalls

- Some common firewall properties:
 - Firewalls are resistant to network attacks.
 - All traffic flows through the firewall.
 - Firewalls enforce the access control policy.
- Several benefits of using a firewall in a network:
 - Prevents the exposure of sensitive hosts, resources, and applications to untrusted users.
 - Sanitizes protocol flow.
 - Blocks malicious data from servers and clients.
 - Reduces security management complexity.
- Firewalls also present some limitations:
 - A misconfigured firewall can have serious consequences for the network.
 - The data from many applications cannot be passed over firewalls securely.
 - Users search for ways around the firewall to receive blocked material.
 - Network performance can slow down.
 - Unauthorized traffic can be tunneled as legitimate traffic through the firewall.

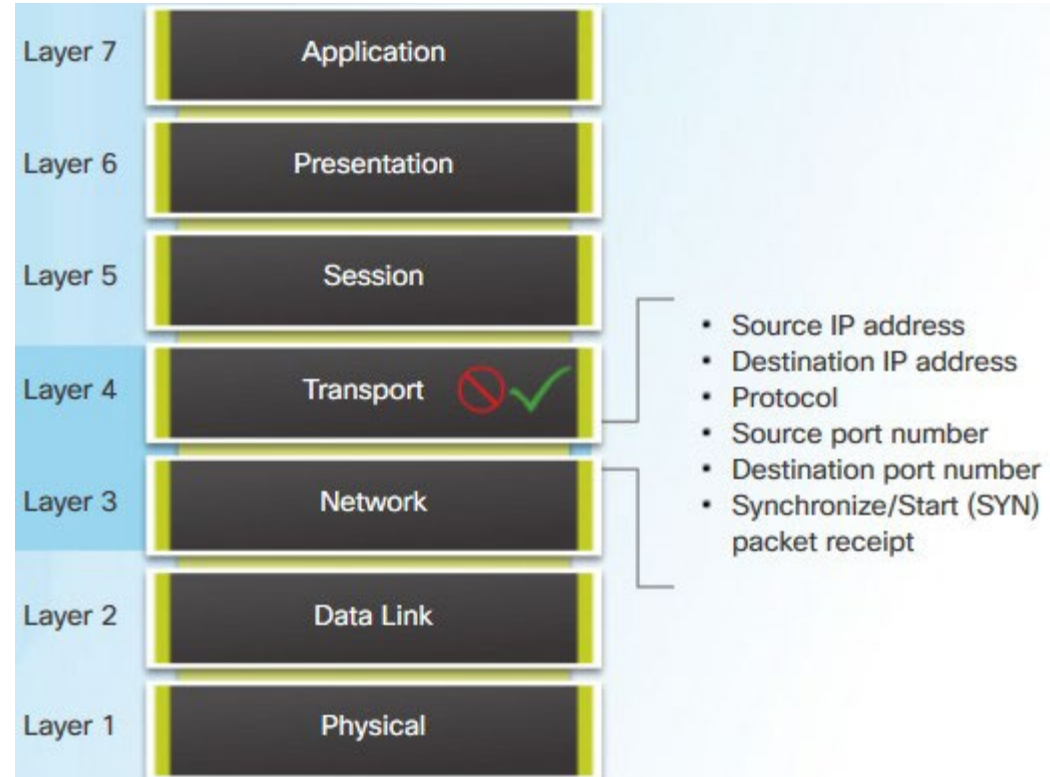


Firewall Type Descriptions

- **Packet filtering (Stateless) firewalls** - usually part of a router firewall, which permits or denies traffic based on Layer 3 and Layer 4 information.
- **Stateful firewalls:**
 - Allows or blocks traffic based on state, port, and protocol.
 - Monitors all activity from the opening of a connection until it is closed.
- **Application gateway firewalls (Proxy firewall)** - filters information at Layers 3, 4, 5, and 7 of the OSI reference model.
- **Host-based (server and personal) firewall** - A PC or server with firewall software running on it.
- **Transparent firewall** - filters IP traffic between a pair of bridged interfaces.
- **Hybrid firewall** - a combination of the various firewall types.

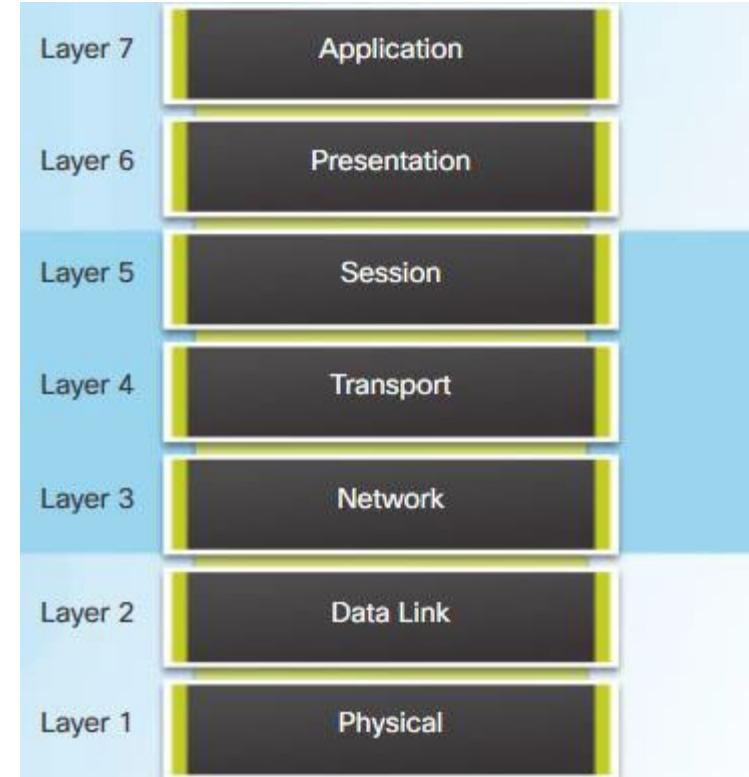
Packet Filtering Firewalls

- Usually part of a router firewall, which permits or denies traffic based on Layer 3 and Layer 4 information.
- Are stateless firewalls that use a simple policy table look-up that filters traffic based on specific criteria.



Stateful Firewalls

- The most versatile and common firewall technology in use.
- Provides stateful packet filtering by using connection information maintained in a state table.
- Classified at the network layer but also analyzes traffic at OSI Layer 4 and Layer 5.



Next-Generation Firewalls

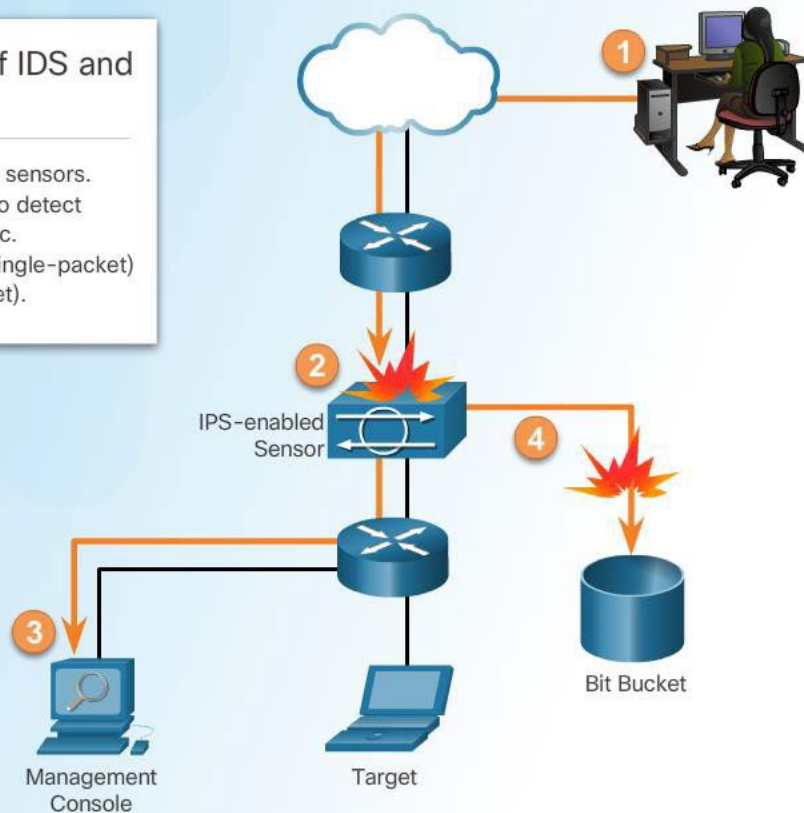
- Provide standard firewall capabilities like stateful inspection.
- Contain integrated intrusion prevention.
- Use application awareness and control to see and block risky apps.
- Upgrade paths to include future information feeds.
- Implement techniques to address evolving security threats.



Intrusion Prevention and Detection Devices

Common Characteristics of IDS and IPS

- Both technologies are deployed as sensors.
- Both technologies use signatures to detect patterns of misuse in network traffic.
- Both can detect atomic patterns (single-packet) or composite patterns (multi-packet).



Advantages and Disadvantages of IDS and IPS

	Advantages	Disadvantages
IDS	▪ No impact on network (latency, jitter) ▪ No network impact if there is a sensor failure ▪ No network impact if there is sensor overload	▪ Response action cannot stop trigger packets ▪ Correct tuning required for response actions ▪ More vulnerable to network security evasion techniques
IPS	▪ Stops trigger packets ▪ Can use stream normalization techniques	▪ Sensor issues might affect network traffic ▪ Sensor overloading impacts the network ▪ Some impact on network (latency, jitter)

Types of IPS

- Host-based IPS (HIPS):
 - Software installed on a single host to monitor and analyze suspicious activity.
 - Monitor and protect operating system and critical system processes that are specific to that host.
 - Combine antivirus software, antimalware software, and firewall.
- Network-based IPS:
 - Implemented using a dedicated or non-dedicated IPS device.
 - Are a critical component of intrusion prevention.
 - Sensors detect malicious and unauthorized activity in real time and can take action when required.

	Advantages	Disadvantages
Host-Based IPS	• Provides protection specific to a host operating system • Provides operating system and application level protection • Protects the host after the message is decrypted	• Operating system dependent • Must be installed on all hosts

Security Devices

Specialized Security Appliances

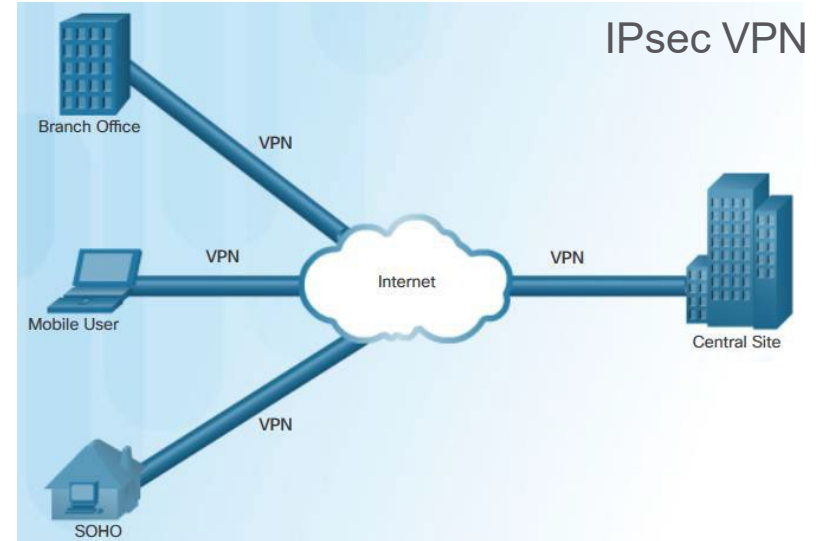
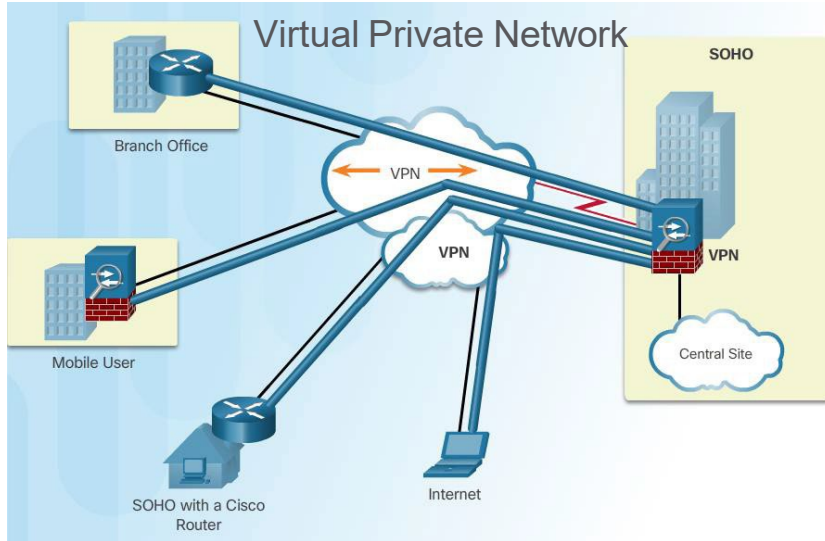
- Cisco Advanced Malware Protection (AMP):
 - Is enterprise-class advanced malware analysis and protection solution.
 - Provides comprehensive malware protection for organizations before, during, and after an attack.
- Cisco Web Security Appliance (WSA) with Cloud Web Security (CWS):
 - WSA protects the network by automatically blocking risky sites and testing unknown sites before allowing users to access them.
 - WSA provides malware protection, application visibility and control, acceptable use policy controls, insightful reporting and secure mobility.
 - CWS enforces secure communication to and from the Internet.
 - CWS provides remote workers the same level of security as onsite employees.
- Cisco Email Security Appliance (ESA):
 - Defends mission-critical email systems.
 - Detects and correlates threats using a worldwide database monitoring system.



Security Services

VPN

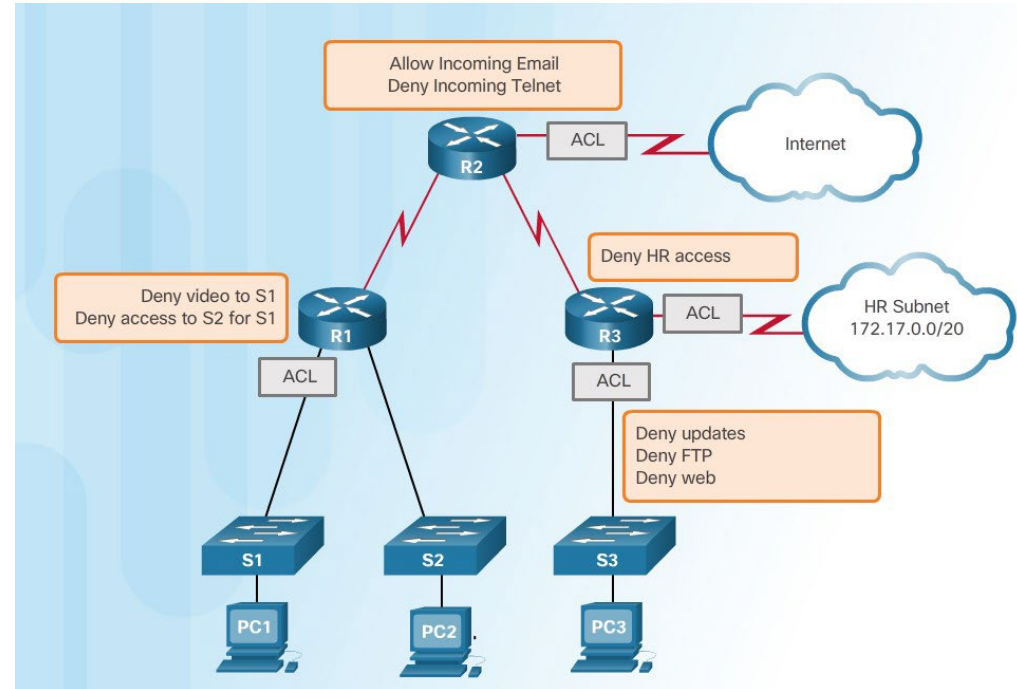
- This is a private network that is created over a public network.
- A VPN is private in that the traffic is encrypted to keep the data confidential while it is transported across the public network.
- IPsec services allow for authentication, integrity, access control, and confidentiality.



Security Services

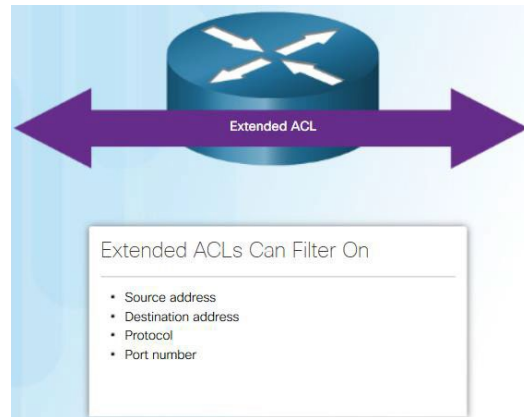
Traffic Control with ACLs

- Access Control Lists (ACLs) - Is a series of commands that control whether a device forwards or drops packets based on information found in the packet header:
 - Limit network traffic to increase network performance.
 - Provide traffic flow control.
 - Provide a basic level of security for network access.
 - Filter traffic based on traffic type.
 - Screen hosts to permit or deny access to network services.



ACLs: Important Features

- The two types of Cisco IPv4 ACLs are standard and extended.
- Standard ACLs can be used to permit or deny traffic only from source IPv4 addresses. Extended ACLs filter IPv4 packets based on several attributes that include:
 - Protocol type
 - Source IPv4 address
 - Destination IPv4 address
 - Source TCP or UDP ports
 - Destination TCP or UDP ports
 - Optional protocol type information for finer control
- Standard and extended ACLs can be created using either a number or a name to identify the ACL and its list of statements.
- An ACL message can be generated and logged when traffic meets the permit or deny criteria defined in the ACL.



Purpose of ACLs

ACL Operation

Inbound and Outbound ACLs



- ACLs do not act on packets that originate from the router itself.
 - ACLs define the set of rules that give added control for packets that enter inbound interfaces, packets that relay through the router, and packets that exit outbound interfaces of the router.
- ACLs can be configured to apply to inbound traffic and outbound traffic:
 - Inbound ACLs – Incoming packets are processed before they are routed to the outbound interface.
 - Outbound ACLs – Incoming packets are routed to the outbound interface, and then they are processed through the outbound ACL.

Wildcard Masks in ACLs

Introducing ACL Wildcard Masking

Wildcard Masking

Octet Bit Position and Address Value for Bit



Examples

0 0 0 0 0 0 0 0 = Match All Address Bits (Match All)

0 0 1 1 1 1 1 1 = Ignore Last 6 Address Bits

0 0 0 0 1 1 1 1 = Ignore Last 4 Address Bits

1 1 1 1 1 1 0 0 = Ignore First 6 Address Bits

1 1 1 1 1 1 1 1 = Ignore All Bits in Octet

0 means to match the value of the corresponding address bit
1 means to ignore the value of the corresponding address bit

- IPv4 ACEs require the use of wildcard masks.
- A wildcard mask is a string of 32 binary digits (1s and 0s) used by the router to determine which bits of the address to examine for a match.
- Wildcard masks are often referred to as an inverse mask since unlike a subnet mask where a binary 1 is a match, a binary 0 is a match with wildcard masks. For example:

	Decimal Address	Binary Address
IP Address to be Processed	192.168.10.0	11000000.10101000.00001010.00000000
Wildcard Mask	0.0.255.255	00000000.00000000.11111111.11111111
Resulting IP Address	192.168.0.0	11000000.10101000.00000000.00000000

Wildcard Masks in ACLs

Wildcard Mask Examples

Wildcard Masks to Match IPv4 Hosts and Subnets

Example 1

	Decimal	Binary
IP Address	192.168.1.1	11000000.10101000.00000001.00000001
Wildcard Mask	0.0.0.0	00000000.00000000.00000000.00000000
Result	192.168.1.1	11000000.10101000.00000001.00000001

Example 2

	Decimal	Binary
IP Address	192.168.1.1	11000000.10101000.00000001.00000001
Wildcard Mask	255.255.255.255	11111111.11111111.11111111.11111111
Result	0.0.0.0	00000000.00000000.00000000.00000000

Example 3

	Decimal	Binary
IP Address	192.168.1.1	11000000.10101000.00000001.00000001
Wildcard Mask	0.0.0.255	00000000.00000000.00000000.11111111
Result	192.168.1.0	11000000.10101000.00000001.00000000

- Calculating the wildcard mask to match IPV4 subnets takes practice. In the first to the left:
 - Example 1: The wildcard mask stipulates that every bit in the IPv4 192.168.1.1 address must match exactly.
 - Example 2: The wildcard mask stipulates that anything will match.
 - Example 3: The wildcard mask stipulates that any host within the 192.168.1.0/24 network will match.

Wildcard Masks in ACLs

Wildcard Mask Keywords

Wildcard Bit Mask Abbreviations

Example 1

- 192.168.10.10 0.0.0.0 matches all of the address bits
- Abbreviate this wildcard mask using the IP address preceded by the keyword **host** (**host 192.168.10.10**)



Example 2

- 0.0.0.0 255.255.255.255 ignores all address bits
- Abbreviate expression with the keyword **any**



- To make wildcard masks easier to read, the keywords **host** and **any** can help identify the most common uses of wildcard masking.
 - **host** substitutes for the 0.0.0.0 mask
 - **any** substitutes for the 255.255.255.255 mask
- If you would like to match the 192.169.10.10 address, you could use **192.168.10.10 0.0.0.0** or, you can use: **host 192.168.10.10**
- In Example 2, instead of entering **0.0.0.0 255.255.255.255**, you can use the keyword **any** by itself.

Wildcard Masks in ACLs

Wildcard Mask Keyword Examples

The any and host Keywords

Example 1

```
R1(config)# access-list 1 permit 0.0.0.0 255.255.255.255
!OR
R1(config)# access-list 1 permit any
```

Example 2

```
R1(config)# access-list 1 permit 192.168.10.10 0.0.0.0
!OR
R1(config)# access-list 1 permit host 192.168.10.10
```

This is the format of the **host** and **any** optional keywords in an ACL statement.

- Example 1 in the figure demonstrates how to use the **any** keyword to substitute the IPv4 address 0.0.0.0 with a wildcard mask of 255.255.255.255.
- Example 2 demonstrates how to use the **host** keyword to substitute for the wildcard mask when identifying a single host.

Guidelines for ACL Creation

General Guidelines for Creating ACLs

ACL Traffic Filtering on a Router



One list per interface, per direction, and per protocol

With two interfaces and two protocols running, this router could have a total of 8 separate ACLs applied.

The Rules for Applying ACLs

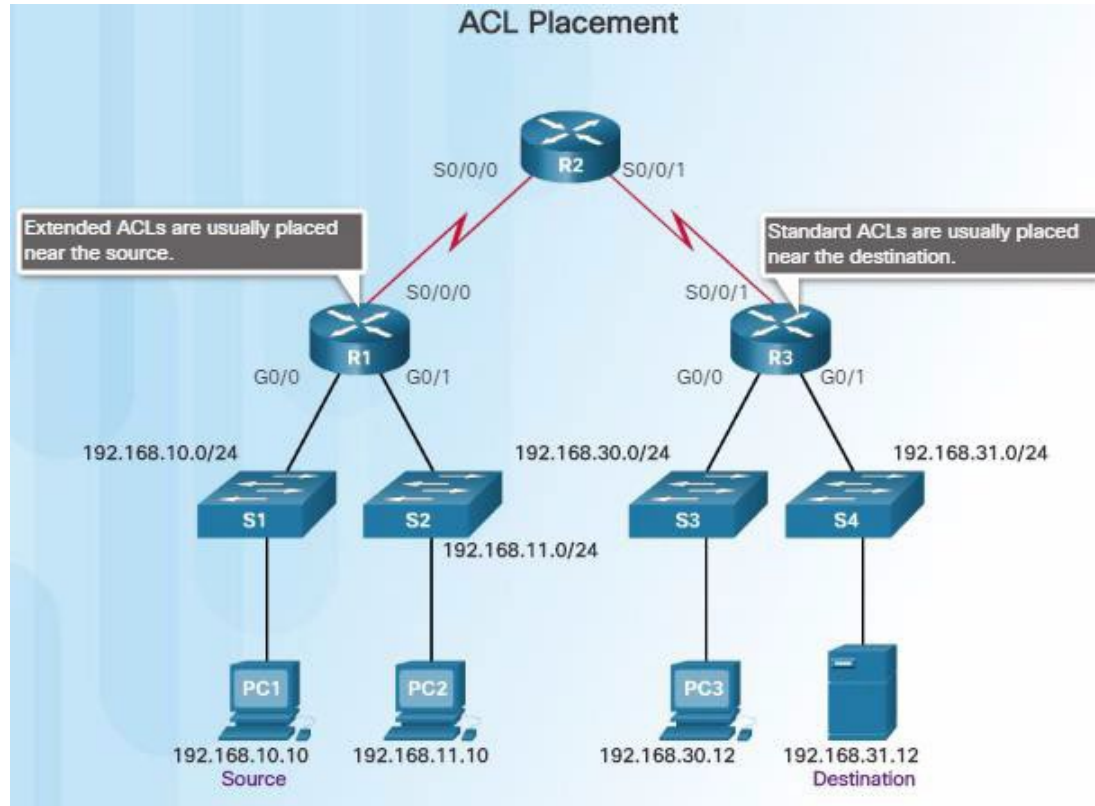
You can only have one ACL per protocol, per interface, and per direction:

- One ACL per protocol (e.g., IPv4 or IPv6)
- One ACL per direction (i.e., IN or OUT)
- One ACL per interface (e.g., GigabitEthernet0/0)

- Use ACLs in firewall routers positioned between your internal network and an external network such as the Internet.
- Use ACLs on a router positioned between two parts of your network to control traffic entering or exiting a specific part of your internal network.
- Configure ACLs on border routers such as those situated at the edge of your network. This will provide a basic buffer from the outside network that is less controlled.
- Configure ACLs for each network protocol configured on the border router interfaces.

Guidelines for ACL Creation

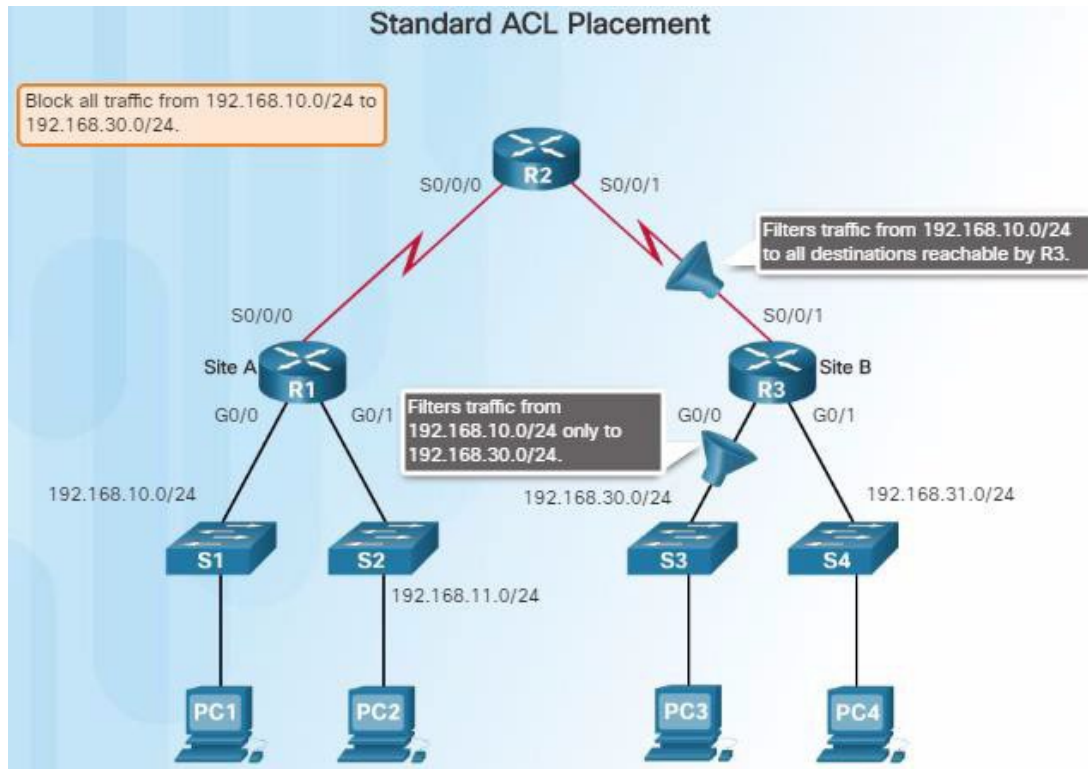
General Guidelines for Creating ACLs



- The proper placement of an ACL can make the network operate more efficiently. For example, an ACL can be placed to reduce unnecessary traffic.
- Every ACL should be placed where it has the greatest impact on efficiency.
 - Extended ACLs – Configure extended ACLs as close as possible to the source of the traffic to be filtered. This will prevent undesirable traffic as close to the source without it crossing the network infrastructure.
 - Standard ACLs – Since standard ACLs do not specify destination addresses, they should be configured as close to the destination as possible.

Guidelines for ACL Creation

Standard ACL Placement



- This example demonstrates the proper placement of the standard ACL that is configured to block traffic from the 192.168.10.0/24 network to the 192.168.30.0/24 network.
- There are two possible places to configure the access-list on R3.
- If the access-list is applied to the S0/0/1 interface, it will block traffic to the 192.168.30.0/24 network, **but also**, going to the 192.168.31.0/24 network.
- The best place to apply the access list is on R3's G0/0 interface. The access-list should be applied to traffic exiting the G0/0 interface. Packets from 192.168.10.0/24 can still reach 192.168.31.0/24.

Configure Standard IPv4 ACLs

Numbered Standard IPv4 ACL Syntax

Parameter	Description
<code>access-list-number</code>	Number of an ACL. This is a decimal number from 1 to 99, or 1300 to 1999 (for standard ACL).
<code>deny</code>	Denies access if the conditions are matched.
<code>permit</code>	Permits access if the conditions are matched.
<code>remark</code>	Add a remark about entries in an IP access list to make the list easier to understand and scan.
<code>source</code>	Number of the network or host from which the packet is being sent. There are two ways to specify the <code>source</code> : • Use a 32-bit quantity in four-part, dotted-decimal format. • Use the keyword any as an abbreviation for a <code>source</code> and <code>source-wildcard</code> of 0.0.0.0 255.255.255.255.
<code>source-wildcard</code>	(Optional) 32-bit wildcard mask to be applied to the source. Places ones in the bit positions you want to ignore.
<code>log</code>	(Optional) Causes an informational logging message about the packet that matches the entry to be sent to the console. (The level of messages logged to the console is controlled by the logging console command.) The message includes the ACL number, whether the packet was permitted or denied, the source address, and the number of packets. The message is generated for the first packet that matches, and then at five-minute intervals, including the number of packets permitted or denied in the prior five-minute interval.

- The **access-list** global configuration command defines a standard ACL with a number in the range of 1 through 99.
- The full syntax of the standard ACL command is as follows:

```
Router(config)# access-list access-list-number { deny | permit | remark } source [ source-wildcard ] [ log ]
```

To remove the ACL, the global configuration **no access-list** command is used. Use the **show access-list** command to verify the removal of the ACL.

Configure Standard IPv4 ACLs

Applying Standard IPv4 ACLs to Interfaces

Step 1: Use the `access-list` global configuration command to create an entry in a standard IPv4 ACL.

```
R1(config)# access-list 1 permit 192.168.10.0 0.0.0.255
```

The example statement matches any address that starts with 192.168.10.x. Use the `remark` option to add a description to your ACL.

Step 2: Use the `interface` configuration command to select an interface to which to apply the ACL.

```
R1(config)# interface serial 0/0/0
```

Step 3: Use the `ip access-group` interface configuration command to activate the existing ACL on an interface.

```
R1(config-if)# ip access-group 1 out
```

This example activates the standard IPv4 ACL 1 on the interface as an outbound filter.

- After a standard IPv4 ACL is configured, it is linked to an interface using the **ip access-group** command in interface configuration mode:

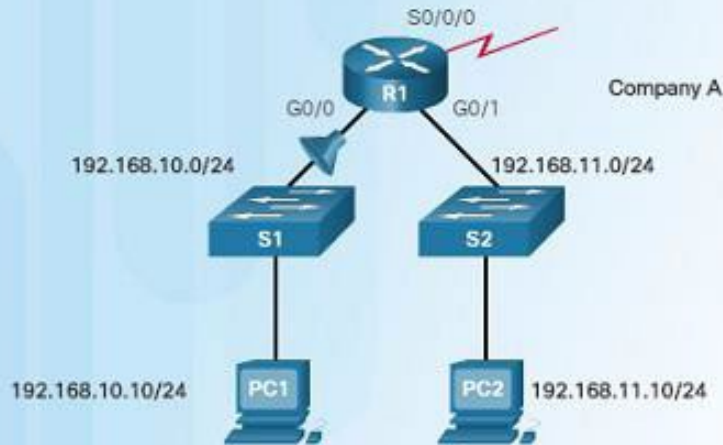
```
Router(config-if)# ip access-group  
{ access-list-number | access-list-name } { in | out }
```

- To remove an ACL from an interface, first enter the **no ip access-group** command on the interface, and then enter the global **no access-list** command to remove the entire ACL.

Configure Standard IPv4 ACLs

Numbered Standard IPv4 ACL Examples

Deny a Specific Host

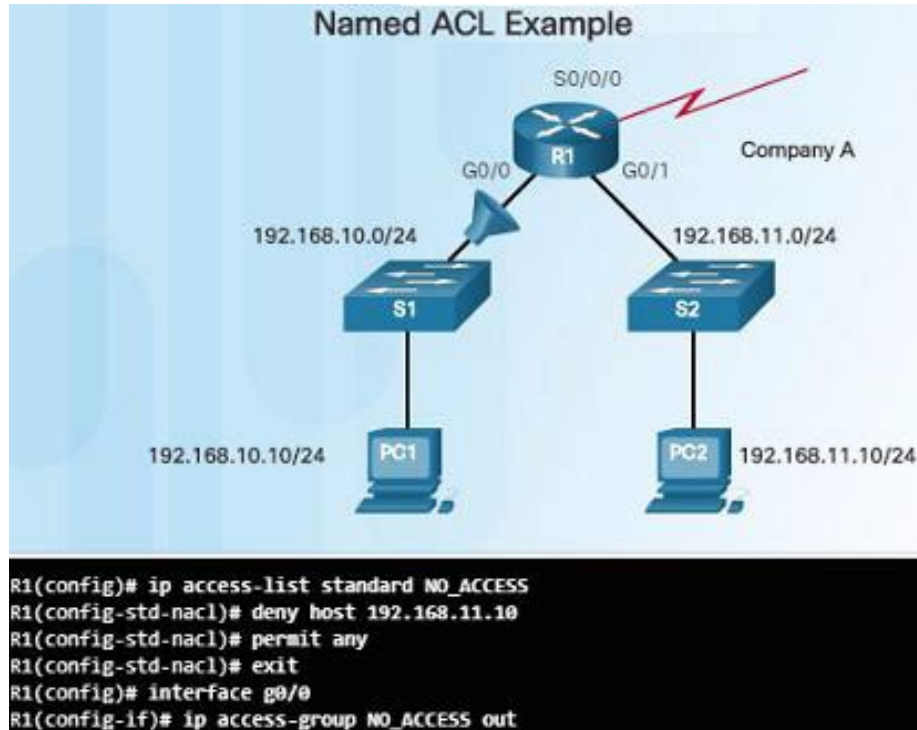


```
R1(config)# no access-list 1
R1(config)# access-list 1 deny host 192.168.10.10
R1(config)# access-list 1 permit any
R1(config)# interface g0/0
R1(config-if)# ip access-group 1 in
```

- This next example demonstrates an ACL that denies a specific host but will permit all other traffic.
 - The first ACL statement deletes the previous version of ACL 1.
 - The next command, with the deny keyword, will deny traffic from the PC1 host that is located at 192.168.10.10.
 - The **access-list 1 permit any** statement will permit all other hosts.
 - This ACL is applied to interface G0/0 in the inbound direction since it only affects the 192.168.10.0/24 LAN.

Configure Standard IPv4 ACLs

Named Standard IPv4 ACL Syntax



- Identifying an ACL with a name rather than with a number makes it easier to understand its function.
- The example to the left shows how to configure a named standard access list. Notice how the commands are slightly different:
 - Use the **ip access-list** command to create a named ACL. Names are alphanumeric, case sensitive, and must be unique.
 - Use permit or deny statements as needed. You can also use the **remark** command to add comments.
 - Apply the ACL to an interface using the **ip access-group name** command.

Modify IPv4 ACLs

Method 1 – Use a Text Editor

Editing Numbered ACLs Using a Text Editor

Configuration

```
R1(config)# access-list 1 deny host 192.168.10.99
R1(config)# access-list 1 permit 192.168.0.0 0.0.255.255
```

Step 1

```
R1# show running-config | include access-list 1
access-list 1 deny host 192.168.10.99
access-list 1 permit 192.168.0.0 0.0.255.255
```

Step 2

<Text editor>

```
access-list 1 deny host 192.168.10.10
access-list 1 permit 192.168.0.0 0.0.255.255
```

Step 3

```
R1# config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# no access-list 1
R1(config)# access-list 1 deny host 192.168.10.10
R1(config)# access-list 1 permit 192.168.0.0 0.0.255.255
```

Step 4

```
R1# show running-config | include access-list 1
access-list 1 deny host 192.168.10.10
access-list 1 permit 192.168.0.0 0.0.255.255
```

- It is sometimes easier to create and edit ACLs in a text editor such as Microsoft Notepad rather making changes directly on the router.
- For an existing ACL, use the **show running-config** command to display the ACL, copy and paste it into the text editor, make the necessary changes, and then paste it back in to the router interface.
- It is important to note that when using the no access-list command, different IOS software releases act differently.
 - If the ACL that has been deleted is still applied to the interface, some IOS versions act as if no ACL is protecting your network while others deny all traffic.

Modify IPv4 ACLs

Method 2 – Use Sequence Numbers

Editing Numbered ACLs Using Sequence Numbers

Configuration

```
R1(config)# access-list 1 deny host 192.168.10.99
R1(config)# access-list 1 permit 192.168.0.0 0.0.255.255
```

Step 1

```
R1# show access-lists 1
Standard IP access list 1
 10 deny 192.168.10.99
 20 permit 192.168.0.0, wildcard bits 0.0.255.255
R1#
```

Step 2

```
R1# conf t
R1(config)# ip access-list standard 1
R1(config-std-nacl)# no 10
R1(config-std-nacl)# 10 deny host 192.168.10.10
R1(config-std-nacl)# end
R1#
```

Step 3

```
R1# show access-lists
Standard IP access list 1
 10 deny 192.168.10.10
 20 permit 192.168.0.0, wildcard bits 0.0.255.255
R1#
```

- The figure to the left demonstrates the steps used to make changes to a numbered ACL using sequence numbers.
- Step 1 identifies the problem. The **deny 192.168.10.99** statement is incorrect. The host to deny should be 192.168.10.10
- To make the edit, Step 2 shows how to go into standard access-list 1 and make the change. The misconfigured statement had to be deleted with the no command: **no 10**
- Once it was deleted, the new statement with the correct host was added: **10 deny host 192.168.10.10**

Modify IPv4 ACLs

Editing Standard Named ACLs

```
R1# show access-lists
Standard IP access list NO_ACCESS
 10 deny 192.168.11.10
 20 permit 192.168.11.0, wildcard bits 0.0.0.255
R1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# ip access-list standard NO_ACCESS
R1(config-std-nacl)# 15 deny host 192.168.11.11
R1(config-std-nacl)# end
R1# show access-lists
Standard IP access list NO_ACCESS
 10 deny 192.168.11.10
 15 deny 192.168.11.11
 20 permit 192.168.11.0, wildcard bits 0.0.0.255
R1#
```

- The **no sequence-number** named ACL command is used to delete individual statements.

- By referring to statement sequence numbers, individual statements can be easily inserted or deleted.
- The figure to the left shows an example of how to insert a line into a named ACL.
- By numbering it 15, it will place the command in between statement 10 and 20.
- Please notice that when the ACL was originally created, the network administrator spaced each command by 10 which left room for edits and additions.

Modify IPv4 ACLs

Verifying ACLs

```
R1# show ip interface s0/0/0
Serial0/0/0 is up, line protocol is up
  Internet address is 10.1.1.1/30
<output omitted>
  Outgoing access list is 1
  Inbound access list is not set
<output omitted>

R1# show ip interface g0/0
GigabitEthernet0/0 is up, line protocol is up
  Internet address is 192.168.10.1/24
<output omitted>
  Outgoing access list is NO_ACCESS
  Inbound access list is not set
<output omitted>
```

```
R1# show access-lists
Standard IP access list 1
  10 deny 192.168.10.10
  20 permit 192.168.0.0, wildcard bits 0.0.255.255
Standard IP access list NO_ACCESS
  15 deny 192.168.11.11
  10 deny 192.168.11.10
  20 permit 192.168.11.0, wildcard bits 0.0.0.255
R1#
```

- Use the **show ip interface** command to verify that the ACL is applied to the correct interface.
- The output will display the name of the access list and the direction in which it was applied to the interface.
- Use the **show access-lists** command to display the access-lists configured on the router.
- Notice how the sequence is displayed out of order for the NO_ACCESS access list. This will be discussed later in this section.

Modify IPv4 ACLs

ACL Statistics

```
R1# show access-lists
Standard IP access list 1
 10 deny 192.168.10.10 (8 match(es))
 20 permit 192.168.0.0, wildcard bits 0.0.255.255
Standard IP access list NO_ACCESS
 15 deny 192.168.11.11
 10 deny 192.168.11.10 (4 match(es))
 20 permit 192.168.11.0, wildcard bits 0.0.0.255
R1# clear access-list counters 1
R1#
R1# show access-lists
Standard IP access list 1
 10 deny 192.168.10.10
 20 permit 192.168.0.0, wildcard bits 0.0.255.255
Standard IP access list NO_ACCESS
 15 deny 192.168.11.11
 10 deny 192.168.11.10 (4 match(es))
 20 permit 192.168.11.0, wildcard bits 0.0.0.255
```

Matches have been
cleared.

- The **show access-lists** command can be used to display matched statistics after an ACL has been applied to an interface and some testing has occurred.
- When traffic is generated that should match an ACL statement, the matches shown in the **show access-lists** command output should increase.
- Recall that every ACL has an implicit **deny any** as the last statement. The statistics for this implicit command will not be displayed. However, if this command is configured manually, the results will be displayed.
- The **clear access-list counters** command can be used to clear the counters for testing purposes.

Processing Packets with ACLs

Cisco IOS Reorders Standard ACLs

Sequencing Considerations During Configuration

```
R1(config)# access-list 1 deny 192.168.10.0 0.0.0.255
R1(config)# access-list 1 deny 192.168.20.0 0.0.0.255
R1(config)# access-list 1 deny 192.168.30.0 0.0.0.255
R1(config)# access-list 1 permit 10.0.0.1
R1(config)# access-list 1 permit 10.0.0.2
R1(config)# access-list 1 permit 10.0.0.3
R1(config)# access-list 1 permit 10.0.0.4
R1(config)# access-list 1 permit 10.0.0.5
R1(config)# end
R1# show running-config | include access-list 1
access-list 1 permit 10.0.0.2
access-list 1 permit 10.0.0.3
access-list 1 permit 10.0.0.1
access-list 1 permit 10.0.0.4
access-list 1 permit 10.0.0.5 access-list 1 deny 192.168.10.0 0.0.0.255
access-list 1 deny 192.168.20.0 0.0.0.255
access-list 1 deny 192.168.30.0 0.0.0.255
R1#
```

Range (network) statements

Host statements

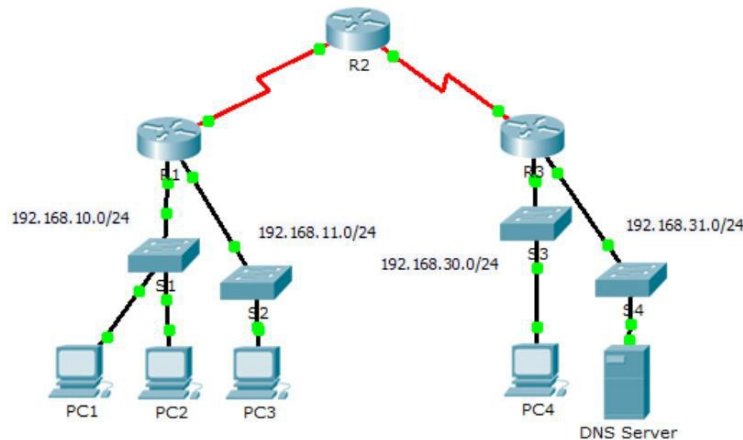
- Note the order in which the access-list statements were entered during configuration.
- Notice how the order was changed when you enter the **show running-config** command.
- The host statements are listed first, however, not in the order they were entered.
- The IOS puts host statements in an order using a special hashing function. The resulting order optimizes the search for a host ACL entry.
- The range statements are displayed in the order they were entered. The hashing function is applied to host statements.

Packet Tracer – ACL Demonstration



Packet Tracer – Access Control List Demonstration

Topology



Objectives

Part 1: Verify Local Connectivity and Test Access Control List

Part 2: Remove Access Control List and Repeat Test

Note!

Extended ACL and more configuration will be discussed in
Internetworking course

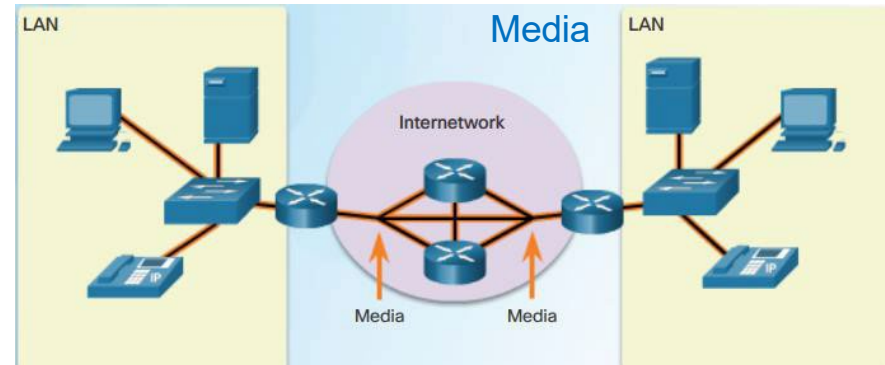
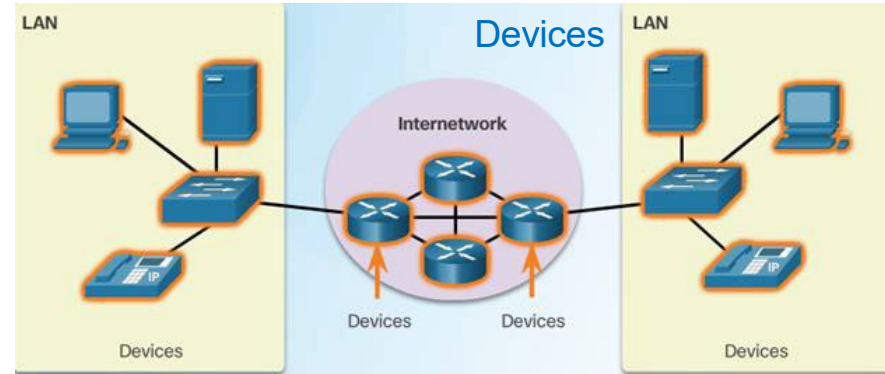
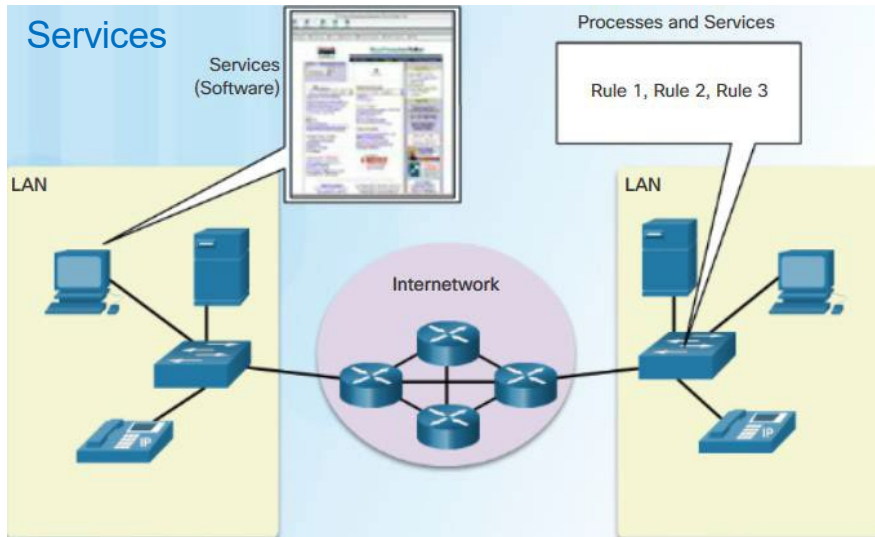
11.2 Network Representations

Network Topologies

Overview of Network Components

- Network infrastructure contains three categories of network components:

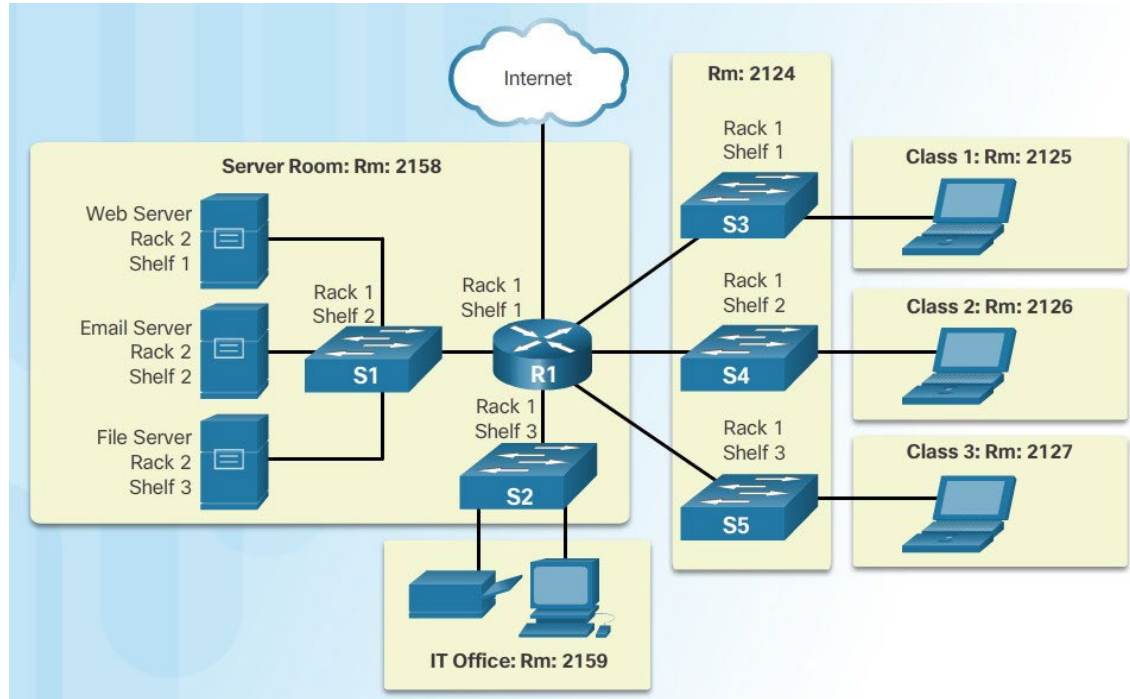
- Devices
- Media
- Services



Network Topologies

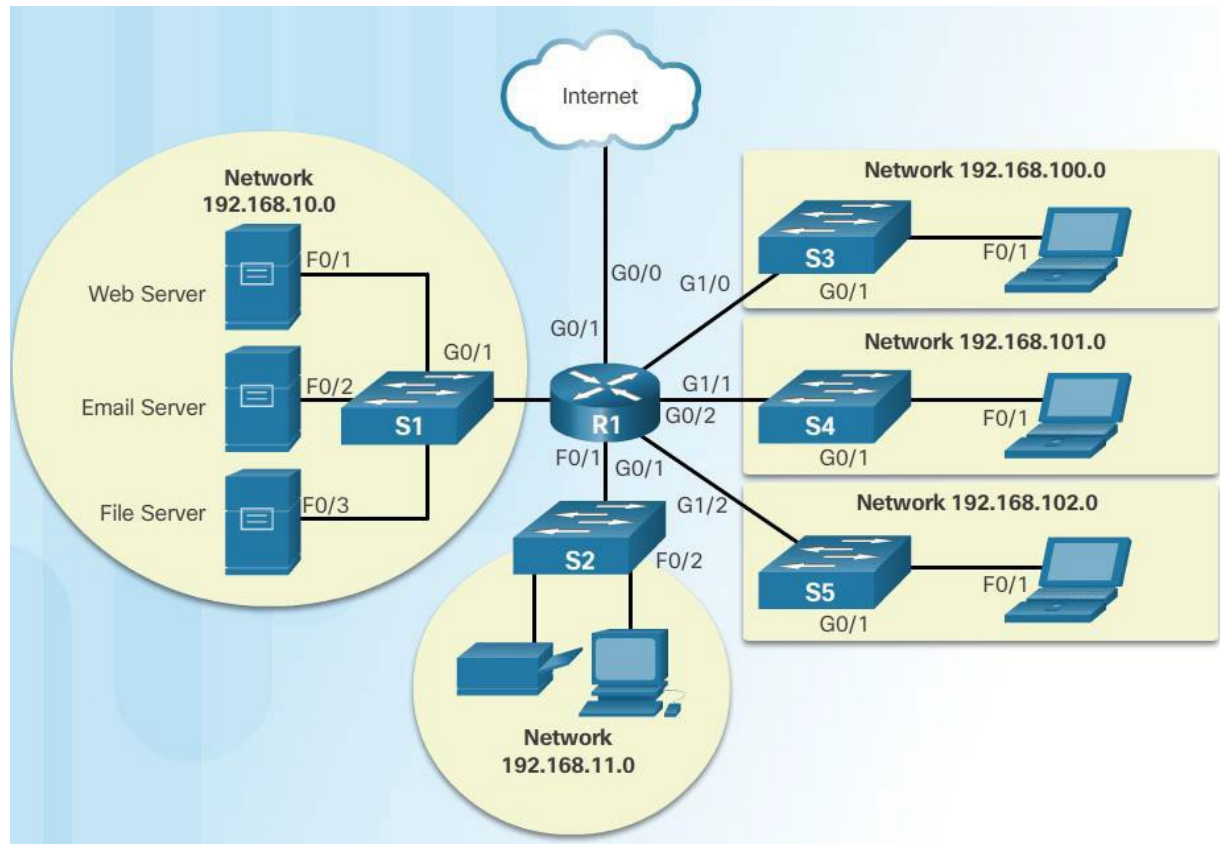
Physical and Logical Topologies

- Physical Topology refers to the physical connections and identifies how end devices and infrastructure devices are interconnected.



Physical and Logical Topologies (Cont.)

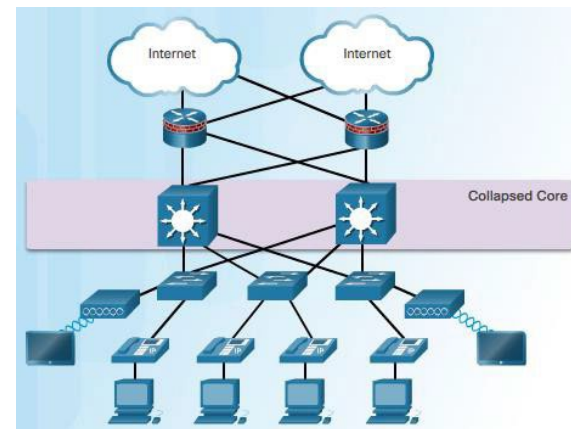
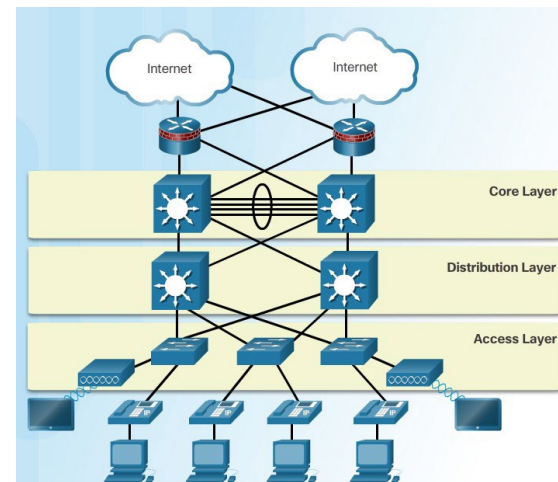
- Logical Topology refers to the way a network transfers frames from one node to the next.



The Three-Layer Network Design Model

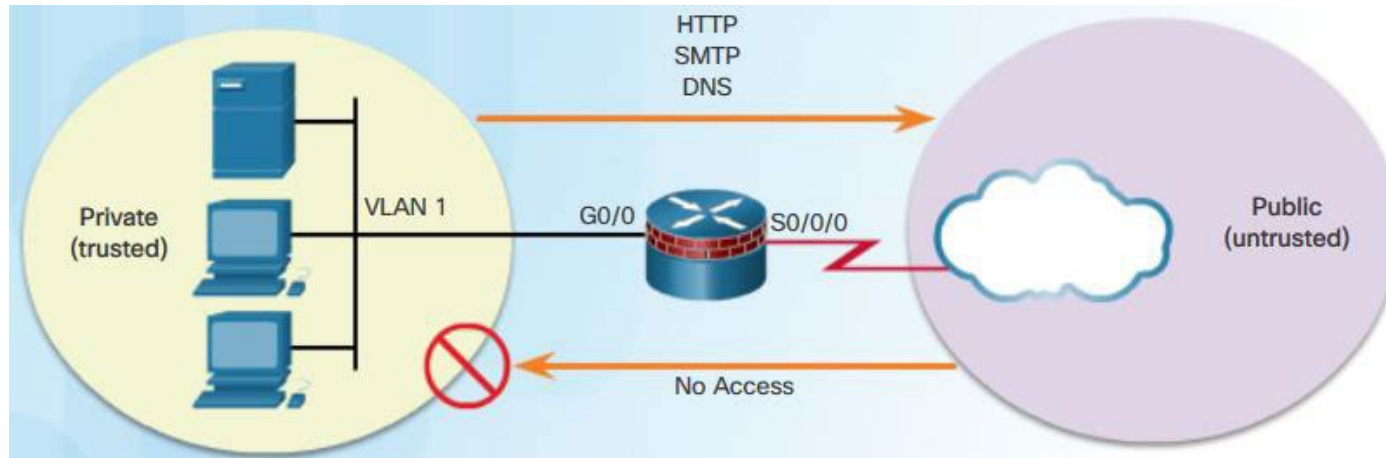
Three-Layer Hierarchical Model

- Access layer:
 - Provides endpoints and users direct access to the network.
 - User traffic is initiated at this layer.
 - Distribution layer
 - Aggregates access layers.
 - Provides connectivity to services.
 - Core layer
 - Provides connectivity between distribution layers.
- Collapsed Core
- Core and distribution layers are collapsed into one layer.
 - Reduces cost and complexity.



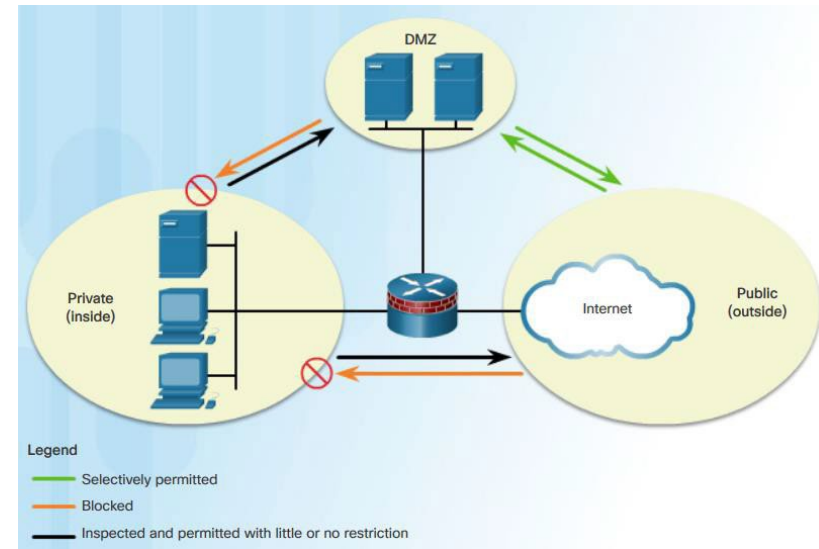
Common Security Architectures

- Firewall design is primarily about device interfaces permitting or denying traffic based on the source, the destination, and the type of traffic. Some designs are as simple as designating an outside network and inside network. A firewall with two interfaces is configured as follows:
 - Traffic originating from the private network is permitted and inspected as it travels toward the public network. Inspected traffic returning from the public network and associated with traffic that originated from the private network is permitted.
 - Traffic originating from the public network and traveling to the private network is generally blocked.



Common Security Architectures (Cont.)

- A demilitarized zone (DMZ) is a firewall design where there is typically one inside interface connected to the private network, one outside interface connected to the public network, and one DMZ interface:
- Traffic originating from the private network is inspected as it travels toward the public or DMZ network. This traffic is permitted with little or no restriction. Return traffic is usually permitted.
- Traffic originating from the DMZ network and traveling to the private network is usually blocked.
- Traffic originating from the DMZ network and traveling to the public network is selectively permitted based on service requirements.
- Traffic originating from the public network and traveling toward the DMZ is selectively permitted and inspected. Return traffic is dynamically permitted.
- Traffic originating from the public network and traveling to the private network is blocked.



Common Security Architectures (Cont.)

- Zone-based policy firewalls (ZPFs) use the concept of zones to provide additional flexibility.
- A zone is a group of one or more interfaces that have similar functions or features.

